

◆ 16.11.5 Evaluation and Decision Scoring Integration (Final — Integrated Version)

Instantaneous Evaluation Function

$$f_i(t) = \overline{\mathcal{E}}(\Psi_i(t)), \nabla \Phi(\Psi_i(t)) - 2\sigma^2 \nabla \ln P_i(t)$$

Under compactness of Ω and bounded gradients:

$$|f_i(t)| \leq C_f$$

Windowed Evaluation

$$G_{i,u}^k = \int_{t_k}^{t_k + \Delta T} f_{i,u}(t) e^{-\lambda(t-t_k)} dt$$

Thus:

$$|G_{i,u}^k| \leq \frac{C_f}{\lambda}$$

Normalization

$$\tilde{G}_{i,u}^k = \frac{G_{i,u}^k}{E_{\{\mathrm{ref}\}}}$$

Thus:

$$|\tilde{G}_{i,u}^k| \leq C$$

PAC-Based Weighting

$$W_i^k = \frac{PAC_i^k}{\sum_j PAC_j^k + \epsilon_w},$$

$$\quad W_i^k \in (0,1), \quad \sum_i W_i^k = 1$$

Thus:

decision aggregation is a convex combination.

Decision Score

$$D_u(k) = \sum_i W_i^k \tilde{G}_{i,u}^k$$

Thus:

$$|D_u(k)| \leq C_D$$

Temporal Smoothness

$$|D_u(k+1) - D_u(k)| \leq L_D \Delta t$$

Thus:

no abrupt changes occur in decision evaluation.

Goal Alignment

$$f_i \subseteq \nabla \Phi(\Psi_i)$$

Thus:

decision scores are aligned with the system objective.

Real-Time Consistency

$$G_{i,u}^k \text{ is computed over } [t_k, t_{k+}]$$

$\Delta T]$

Thus:

no temporal mismatch occurs between state, probability, and evaluation.

Stability Compatibility

Since:

- f_i is bounded
- weights W_i are bounded and normalized
- scores D_u are bounded

Thus:

the evaluation layer preserves Lyapunov stability and does not introduce destabilizing effects.

Implication

$\boxed{\text{The evaluation layer produces bounded,$

$\text{smooth, and objective-aligned decision scores, fully consistent with system stability.}}$

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**SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
DOI: 10.5281/zenodo.19521219"**